From Raw Listings to a Monitored Model

An end-to-end MLflow pipeline for house-price prediction

Project 4 · Capstone · Elective II: MLOps · Dr. Santosh Kumar

Group 4 — Swagata Bhowmik (B053) · Sumit Kothari (B062)

What Success Means

A real-estate start-up needs automatic, safe house-price estimates.

± ₹5.74 lakh

average error — about 9% of a typical ₹62-lakh listing

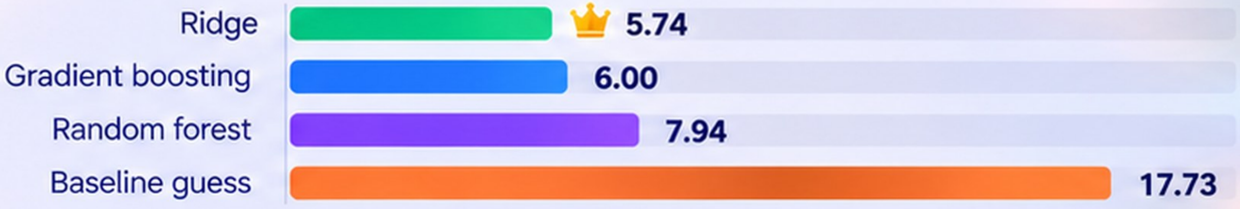
Error (MAE) ≤ ₹7 L
— achieved 5.74

RMSE ≤ ₹9 L
— achieved 7.52

New model ≤ 2% worse than live
— or rejected

p95 ≤ 200 ms · errors ≤ 1%

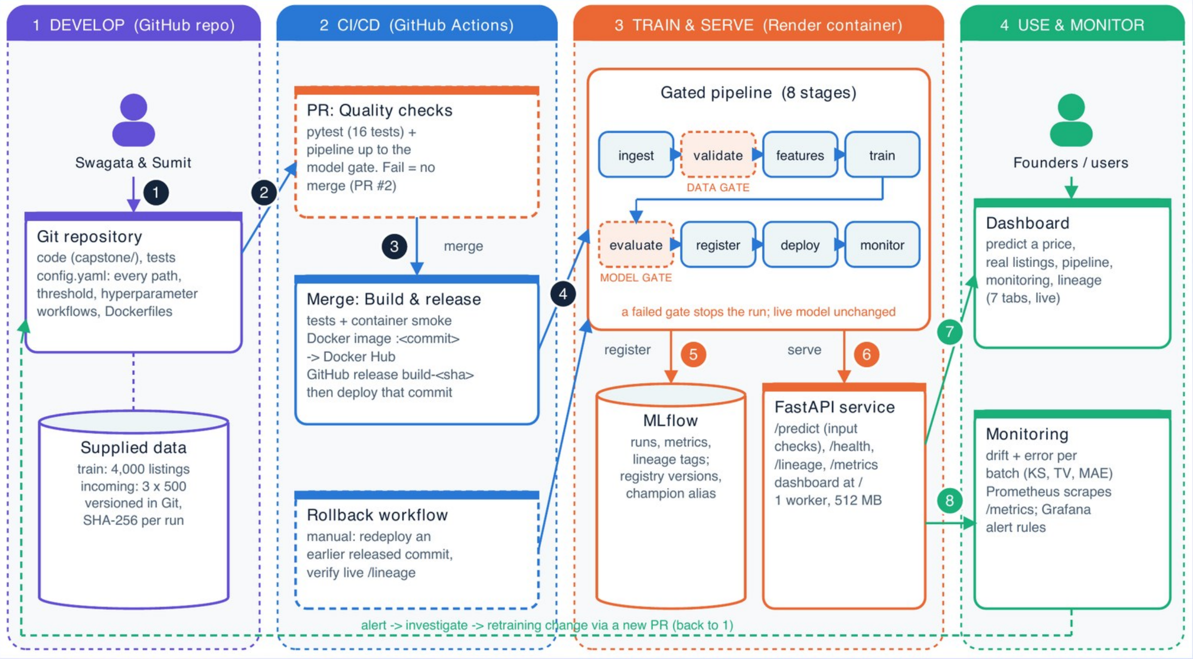
Model comparison (₹ lakh)



Held-out test: MAE 5.78 · RMSE 7.51

Architecture

1 Develop · GitHub
2 CI/CD · GitHub Actions
3 Train & Serve · Render
4 Use & Monitor
alerts → new change



1 DEVELOP (GitHub repo)

Swagata & Sumit

①

Git repository
code (capstone/), tests
config.yaml: every path, threshold, hyperparameter
workflows, Dockerfiles

Supplied data
train: 4,000 listings
incoming: 3 x 500
versioned in Git, SHA-256 per run

2 CI/CD (GitHub Actions)

②

PR: Quality checks
pytest (16 tests) + pipeline up to the model gate. Fail = no merge (PR #2)

③ merge

Merge: Build & release
tests + container smoke
Docker image :<commit> -> Docker Hub
GitHub release build-<sha>
then deploy that commit

Rollback workflow
manual: redeploy an earlier released commit, verify live /lineage

④

3 TRAIN & SERVE (Render container)

Gated pipeline (8 stages)
ingest → validate → features → train
DATA GATE
evaluate → register → deploy → monitor
MODEL GATE
a failed gate stops the run; live model unchanged

register ⑤

serve ⑥

MLflow
runs, metrics, lineage tags; registry versions, champion alias

FastAPI service
/predict (input checks), /health, /lineage, /metrics
dashboard at /
1 worker, 512 MB

4 USE & MONITOR

Founders / users

⑦

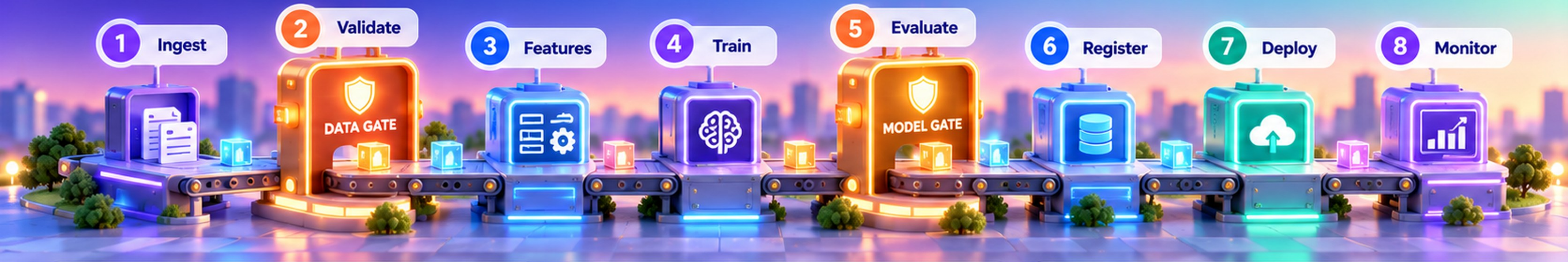
Dashboard
predict a price, real listings, pipeline, monitoring, lineage
(7 tabs, live)

⑧

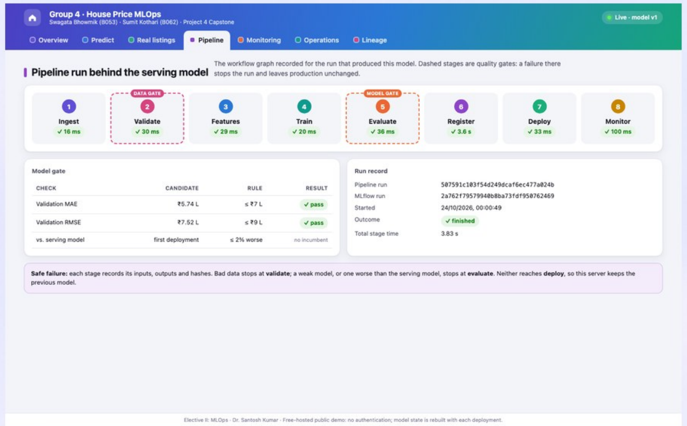
Monitoring
drift + error per batch (KS, TV, MAE)
Prometheus scrapes /metrics; Grafana alert rules

alert -> investigate -> retraining change via a new PR (back to 1)

8 Stages, 2 Safety Gates — Every Run Recorded


1 Ingest
2 Validate
DATA GATE
3 Features
4 Train
5 Evaluate
MODEL GATE
6 Register
7 Deploy
8 Monitor

Workflow graph of the live run


Group 4 · House Price MLOps
Swagata Bhowmik (B053) · Sumit Kothari (B062) · Project 4 Capstone
Live · model v1
Overview
Predict
Real listings
Pipeline
Monitoring
Operations
Lineage

Pipeline run behind the serving model The workflow graph recorded for the run that produced this model. Dashed stages are quality gates: a failure there stops the run and leaves production unchanged.

1 Ingest	2 Validate (DATA GATE)	3 Features	4 Train	5 Evaluate (MODEL GATE)	6 Register	7 Deploy	8 Monitor
✓ 16 ms	✓ 30 ms	✓ 29 ms	✓ 20 ms	✓ 36 ms	✓ 3.6 s	✓ 33 ms	✓ 100 ms

Model gate

CHECK	CANDIDATE	RULE	RESULT
Validation MAE	₹5.74 L	≤ ₹7 L	✓ pass
Validation RMSE	₹7.52 L	≤ ₹9 L	✓ pass
vs. serving model	first deployment	≤ 2% worse	no incumbent

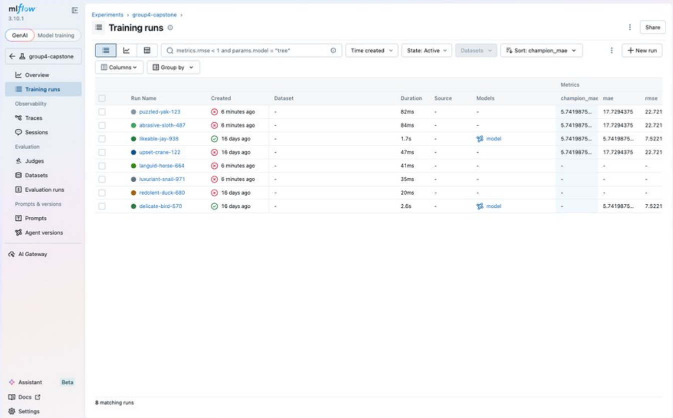
Run record

Pipeline run	507591c103f54d249dcaf6ec477a024b
MLflow run	2a762f79579940b8ba73fdf950762469
Started	24/10/2026, 00:00:49
Outcome	✓ finished
Total stage time	3.83 s

Safe failure: each stage records its inputs, outputs and hashes. Bad data stops at **validate**; a weak model, or one worse than the serving model, stops at **evaluate**. Neither reaches **deploy**, so this server keeps the previous model.

Elective II: MLOps · Dr. Santosh Kumar · Free-hosted public demo: no authentication; model state is rebuilt with each deployment.

Every run recorded, failed ones too


mlflow
3.10.1
GenAI
Model training
group4-capstone
Overview
Training runs
Observability
Traces
Sessions
Evaluation
Judges
Datasets
Evaluation runs
Prompts & versions
Prompts
Agent versions
AI Gateway
Assistant
Beta
Docs
Settings
Experiments › group4-capstone ›
Share
Time created
State: Active
Datasets
Sort: champion_mae
New run
Columns
Group by

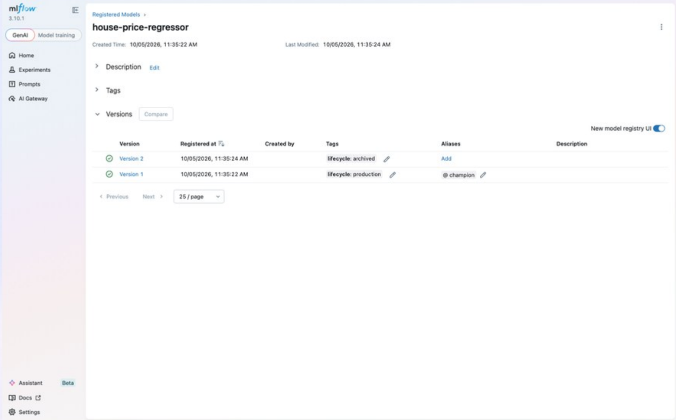
Training runs

metrics.rmse < 1 and params.model = "tree"

Run Name	Created	Dataset	Duration	Source	Models	champion_mae	mae	rmse
puzzled-yak-123	6 minutes ago	-	82ms	-	-	5.7419875...	17.7294375	22.721
abrasive-sloth-487	6 minutes ago	-	84ms	-	-	5.7419875...	17.7294375	22.721
likeable-jay-938	16 days ago	-	1.7s	-	model	5.7419875...	5.7419875...	7.5221
upset-crane-122	16 days ago	-	47ms	-	-	5.7419875...	17.7294375	22.721
languid-horse-664	6 minutes ago	-	41ms	-	-	-	-	-
luxuriant-snail-971	6 minutes ago	-	35ms	-	-	-	-	-
redolent-duck-680	16 days ago	-	20ms	-	-	-	-	-
delicate-bird-570	16 days ago	-	2.6s	-	model	-	5.7419875...	7.5221

8 matching runs

v1 = production + 👑 champion · v2 = archived


mlflow
3.10.1
GenAI
Model training
Home
Experiments
Prompts
AI Gateway
Assistant
Beta
Docs
Settings
Registered Models ›
Description
Edit
Tags
Compare
New model registry UI
Previous
Next
25 / page

house-price-regressor

Created Time: 10/05/2026, 11:35:22 AM

Last Modified: 10/05/2026, 11:35:24 AM

Versions

Version	Registered at	Created by	Tags	Aliases	Description
Version 2	10/05/2026, 11:35:24 AM		lifecycle: archived	Add	
Version 1	10/05/2026, 11:35:22 AM		lifecycle: production	@ champion	

✓ each stage re-runnable

✓ metrics & fingerprints logged

✓ only the pipeline crowns a champion

Two Failures, Stopped Safely

✗ Bad data — stopped at DATA GATE

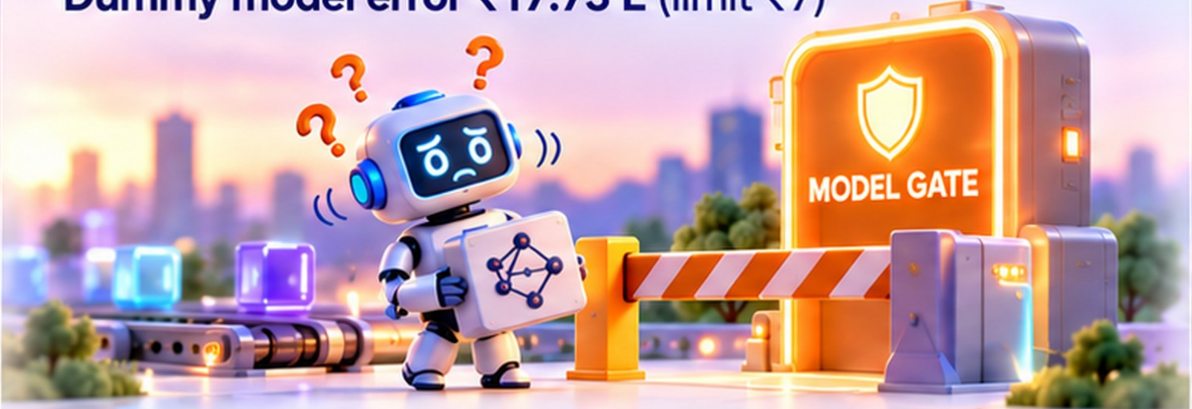
20% of 'Age_Years' blanked (limit 8% missing)

● 1 — ☒ 2 — ○ 3 — ○ 4 — ○ 5 — ○ 6 — ○ 7 — ○ 8

```
Age_Years: missing fraction exceeds limit
```

✗ Bad model — stopped at MODEL GATE

Dummy model error ₹17.73 L (limit ₹7)

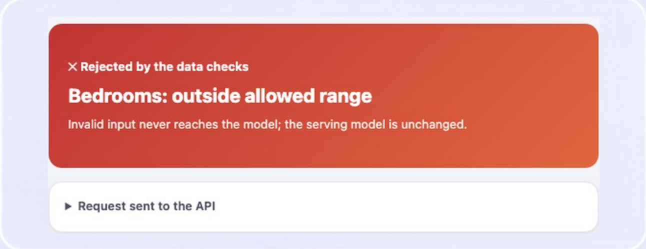


● 1 — ● 2 — ● 3 — ● 4 — ☒ 5 — ○ 6 — ○ 7 — ○ 8

```
Candidate exceeds absolute error thresholds
```

✓ Live server kept serving throughout — 115 requests · 0 errors · model v1 unchanged

Invalid input rejected live



✕ Rejected by the data checks

Bedrooms: outside allowed range

Invalid input never reaches the model; the serving model is unchanged.

▶ Request sent to the API

Where Did the Live Model Come From?

Deployed model — version 1

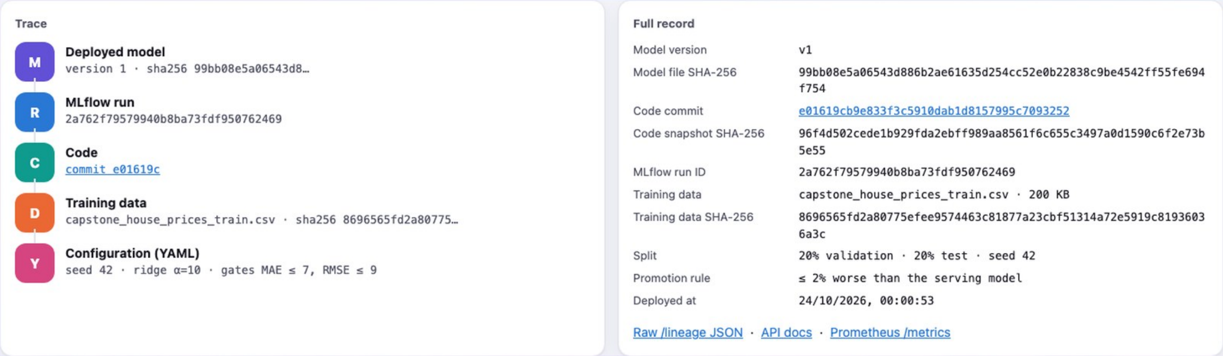
MLflow run — unique run ID

Code — exact Git commit

Training data — SHA-256 fingerprint

Configuration — seed 42 · Ridge α = 10

Live dashboard → Lineage tab



Trace

M **Deployed model**
version 1 · sha256 99bb08e5a06543d8…

R **MLflow run**
2a762f79579940b8ba73fdf950762469

C **Code**
commit e01619c

D **Training data**
capstone_house_prices_train.csv · sha256 8696565fd2a80775…

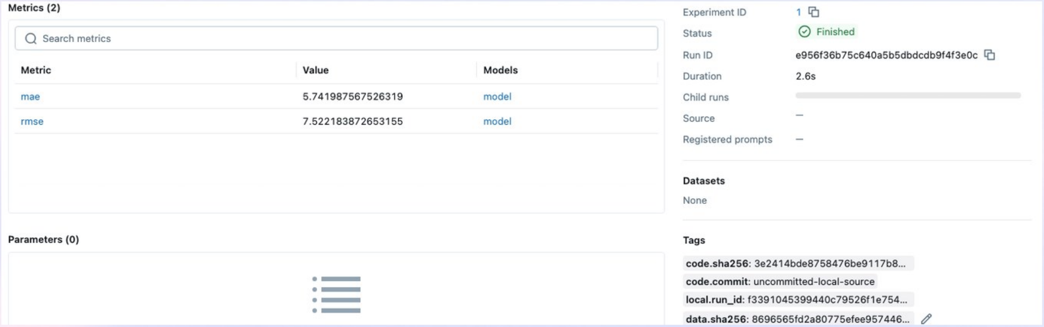
Y **Configuration (YAML)**
seed 42 · ridge α=10 · gates MAE ≤ 7, RMSE ≤ 9

Full record

Model version	v1
Model file SHA-256	99bb08e5a06543d886b2ae61635d254cc52e0b22838c9be4542ff55fe694f754
Code commit	e01619cb9e833f3c5910dab1d8157995c7093252
Code snapshot SHA-256	96f4d502cede1b929fda2ebff989aa8561f6c655c3497a0d1590c6f2e73b5e55
MLflow run ID	2a762f79579940b8ba73fdf950762469
Training data	capstone_house_prices_train.csv · 200 KB
Training data SHA-256	8696565fd2a80775efee9574463c81877a23cbf51314a72e5919c81936036a3c
Split	20% validation · 20% test · seed 42
Promotion rule	≤ 2% worse than the serving model
Deployed at	24/10/2026, 00:00:53

Raw /lineage JSON · API docs · Prometheus /metrics

Same lineage in MLflow (local run): MAE 5.74 · code, commit & data tags



Metrics (2)

Metric	Value	Models
mae	5.741987567526319	model
rmse	7.522183872653155	model

Parameters (0)

Experiment ID	1
Status	Finished
Run ID	e956f36b75c640a5b5dbdcdb9f4f3e0c
Duration	2.6s
Child runs	
Source	—
Registered prompts	—

Datasets

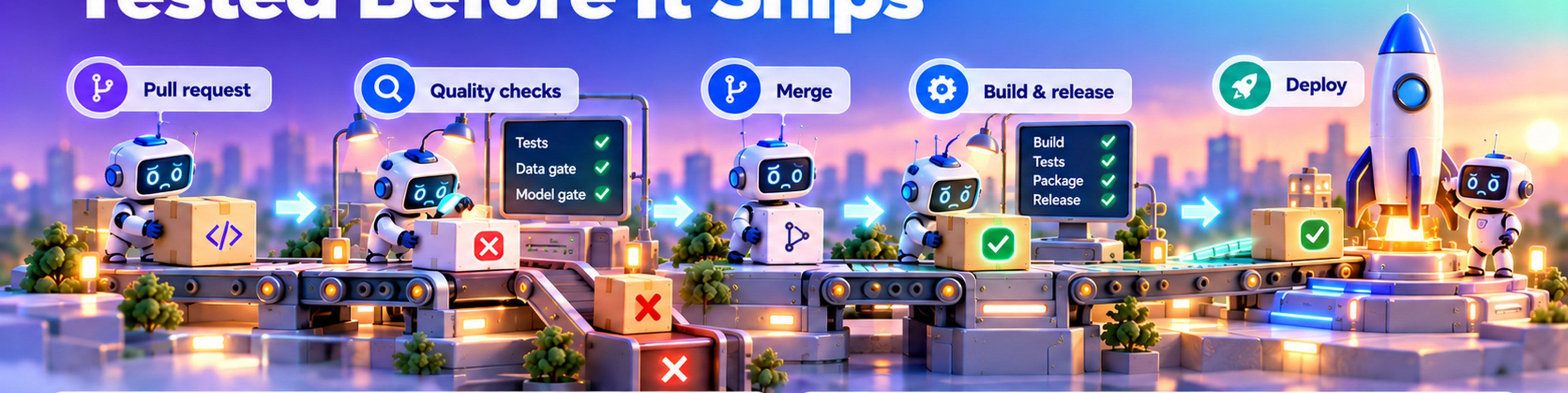
None

Tags

code.sha256: 3e2414bde8758476be9117b8…
code.commit: uncommitted-local-source
local.run_id: f3391045399440c79526f1e754…
data.sha256: 8696565fd2a80775efee957446…

Shown live at group4-mlflow-capstone.onrender.com

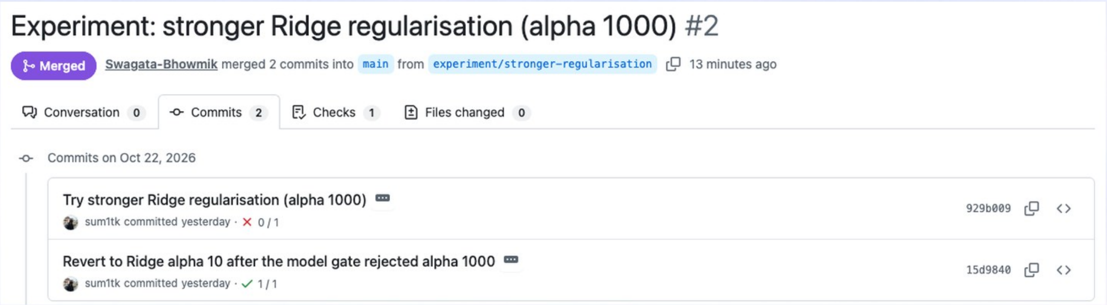
Tested Before It Ships


Pull request
Quality checks
Tests ✓
Data gate ✓
Model gate ✓
Merge
Build & release
Build ✓
Tests ✓
Package ✓
Release ✓
Deploy

PR #2 — a bad change was blocked

- ✗ Ridge α = 1000 → error ₹8.77 L > 7 → CI failed
- ✓ Back to α = 10 → error ₹5.74 → CI passed → merged

by Sumit Kothari



Experiment: stronger Ridge regularisation (alpha 1000) #2

Merged

Swagata-Bhowmik merged 2 commits into `main` from `experiment/stronger-regularisation` 13 minutes ago

Conversation 0
Commits 2
Checks 1
Files changed 0

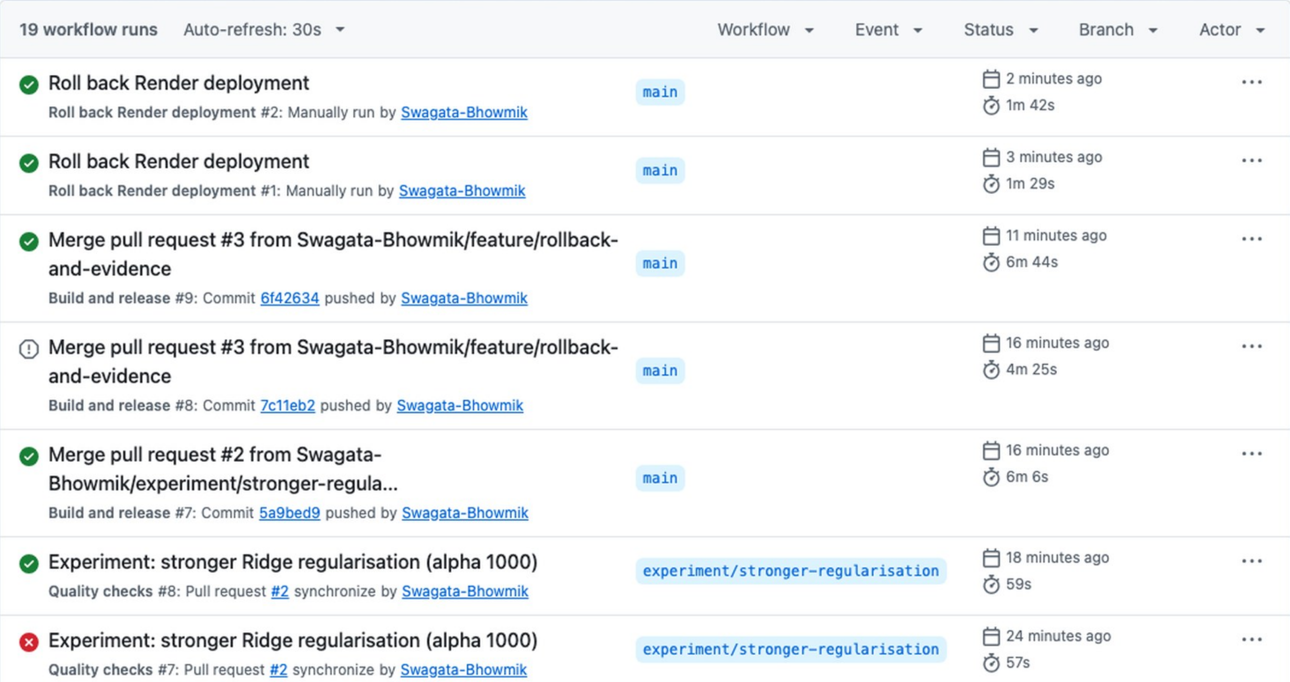
Commits on Oct 22, 2026

Try stronger Ridge regularisation (alpha 1000)
sum1tk committed yesterday · ✗ 0 / 1 — 929b009

Revert to Ridge alpha 10 after the model gate rejected alpha 1000
sum1tk committed yesterday · ✓ 1 / 1 — 15d9840

Release, Rollback & Reproducible

Every merge: tests → Docker image tagged by commit → GitHub release → auto-deploy



19 workflow runs · Auto-refresh: 30s · Workflow · Event · Status · Branch · Actor

- ✓ Roll back Render deployment — `main` — 2 minutes ago · 1m 42s
  Roll back Render deployment #2: Manually run by Swagata-Bhowmik
- ✓ Roll back Render deployment — `main` — 3 minutes ago · 1m 29s
  Roll back Render deployment #1: Manually run by Swagata-Bhowmik
- ✓ Merge pull request #3 from Swagata-Bhowmik/feature/rollback-and-evidence — `main` — 11 minutes ago · 6m 44s
  Build and release #9: Commit 6f42634 pushed by Swagata-Bhowmik
- ! Merge pull request #3 from Swagata-Bhowmik/feature/rollback-and-evidence — `main` — 16 minutes ago · 4m 25s
  Build and release #8: Commit 7c11eb2 pushed by Swagata-Bhowmik
- ✓ Merge pull request #2 from Swagata-Bhowmik/experiment/stronger-regula... — `main` — 16 minutes ago · 6m 6s
  Build and release #7: Commit 5a9bed9 pushed by Swagata-Bhowmik
- ✓ Experiment: stronger Ridge regularisation (alpha 1000) — `experiment/stronger-regularisation` — 18 minutes ago · 59s
  Quality checks #8: Pull request #2 synchronize by Swagata-Bhowmik
- ✗ Experiment: stronger Ridge regularisation (alpha 1000) — `experiment/stronger-regularisation` — 24 minutes ago · 57s
  Quality checks #7: Pull request #2 synchronize by Swagata-Bhowmik

✓ **Live: new version** → **Rolled back · 2 min · old version live** → **Rolled forward · 2 min · new version back**

Each step verified on the live site

Repo:
github.com/Swagata-Bhowmik/group4-mlflow-capstone

Fresh clone:
16/16 tests pass, result identical

One config.yaml — no hard-coded settings

Is the Model Still Healthy?

✓ **Stable**

₹5.56 L

Batch 1 · no drift

⚠ **Data drift**

₹6.87 L

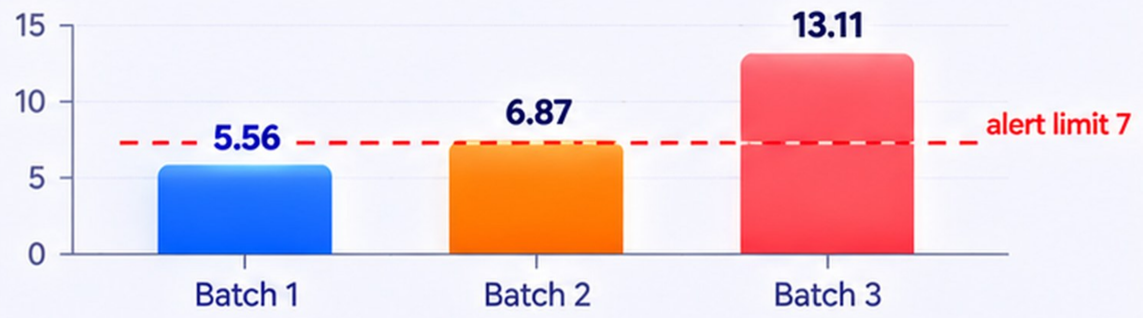
Batch 2 · houses got smaller

✕ **Alert**

₹13.11 L

Batch 3 · error doubled · area, bedrooms, bathrooms, city zone shifted

Average error per batch (₹ lakh)

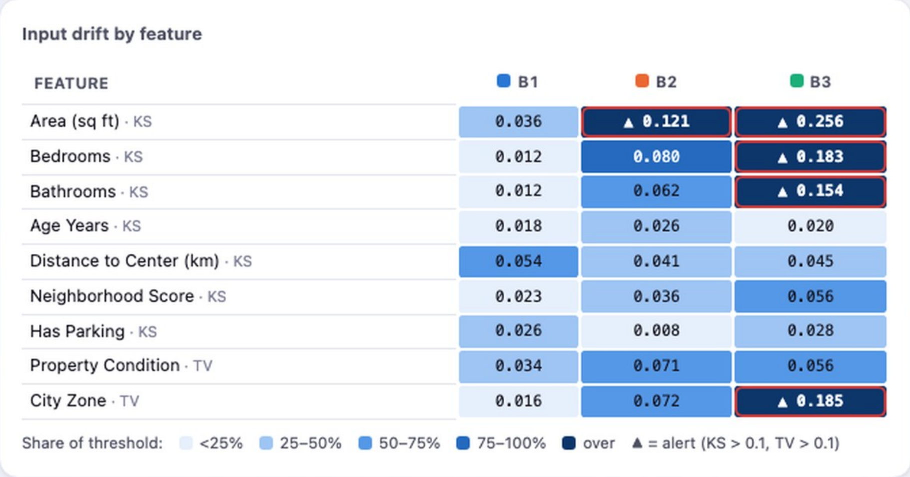


Alert → Keep live model → Investigate → Retrain through all gates

Never auto-replace.

Drift & Live Operations

Drift per feature (alert above 0.10)

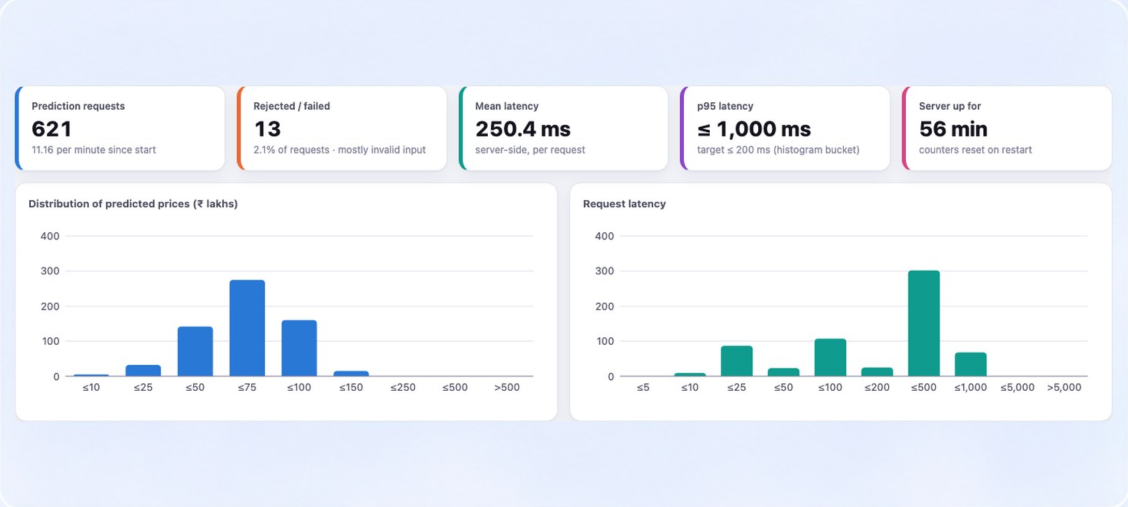


Input drift by feature

FEATURE	■ B1	■ B2	■ B3
Area (sq ft) · KS	0.036	▲ 0.121	▲ 0.256
Bedrooms · KS	0.012	0.080	▲ 0.183
Bathrooms · KS	0.012	0.062	▲ 0.154
Age Years · KS	0.018	0.026	0.020
Distance to Center (km) · KS	0.054	0.041	0.045
Neighborhood Score · KS	0.023	0.036	0.056
Has Parking · KS	0.026	0.008	0.028
Property Condition · TV	0.034	0.071	0.056
City Zone · TV	0.016	0.072	▲ 0.185

Share of threshold: ■ <25% ■ 25–50% ■ 50–75% ■ 75–100% ■ over ▲ = alert (KS > 0.1, TV > 0.1)

Live latency, errors & predicted prices



Prediction requests	Rejected / failed	Mean latency	p95 latency	Server up for
621	**13**	**250.4 ms**	**≤ 1,000 ms**	**56 min**
11.16 per minute since start	2.1% of requests · mostly invalid input	server-side, per request	target ≤ 200 ms (histogram bucket)	counters reset on restart

Distribution of predicted prices (₹ lakhs)

Request latency

✓ **300 live requests · 0 errors**

p95 213 ms · 1 user

⚠ **p95 857 ms · 5 users — above 200 ms on free hosting**

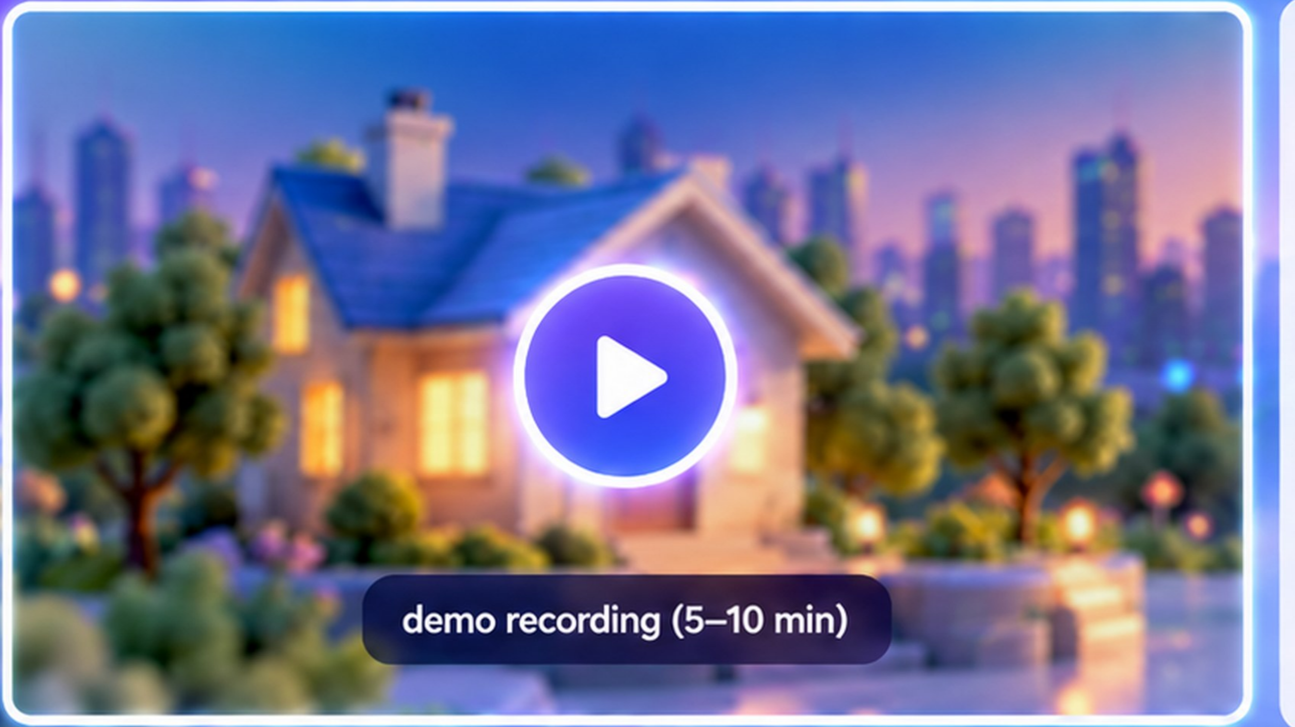
What We'd Improve

What worked: every release decision is automatic, explainable and traceable.

- More serving power — meet the 200 ms target
- Persistent model store — compare every release
- Safer releases — canary + auto-rollback
- Smarter alerts — calibrated, per-zone, notified
- Scale tooling — DVC, Prefect/ZenML

Lesson: data drift and model failure are related — but not the same.

Live Demo & Thank You



demo recording (5–10 min)

In the demo:

1. Code change → pull request
2. CI checks → merge → deploy
3. Pipeline & lineage on the dashboard
4. Monitoring of 3 batches
5. A failure handled safely

Thank You

group4-mlflow-capstone.onrender.com

github.com/Swagata-Bhowmik/group4-mlflow-capstone

Questions?

Group 4 — Swagata Bhowmik (B053) · Sumit Kothari (B062)